Day 163: Confidence Intervals and the Bootstrap, with Real Resampling

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 24, Day 3 | Complete study guide

Watch: Confidence Intervals, Clearly Explained!!! (StatQuest with Josh Starmer) | The standard error, Clearly Explained!!! (StatQuest with Josh Starmer)

Script: `day163_confidence_intervals_bootstrap.py` · every number in this guide is copied by program from `day163_run_output.txt`, the real output of that script.

A 95% confidence interval promises to contain the true value 95% of the time. Today checks that promise instead of trusting it: one whole group of Hillstrom's customers plays the population, thousands of samples are drawn from it, and every interval is scored against the known truth. For a rare purchase rate the textbook interval covered the truth only 0.429 of the time at 100 customers; Wilson's interval held. For spend, which is almost all zeros, no interval reached 95% until 5,000 customers. The bootstrap gives intervals where no formula exists, and paired resampling halves the width of a model comparison (0.0126 against 0.0242). It also fails where it should: an interval for the largest value covered the truth in only 0.030 of samples.

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

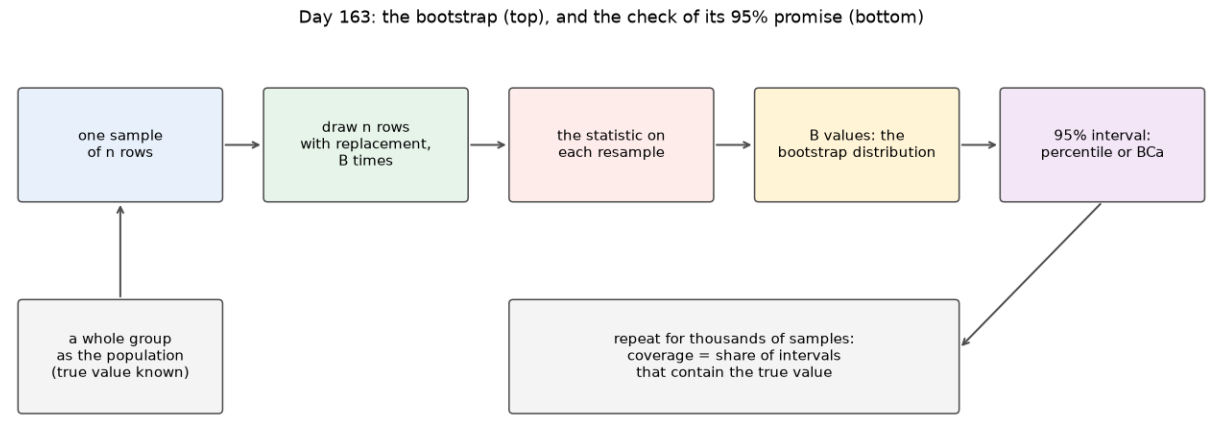
A **confidence interval** is a range computed from a sample. A 95% interval comes from a recipe that, used on many samples, contains the true value 95% of the time; that share is the recipe's **coverage**. Any one interval either contains the truth or does not.

The **bootstrap** estimates how a statistic would vary from sample to sample using only the one sample in hand: draw n rows from it *with replacement* (some rows twice, some not at all), recompute the statistic, repeat thousands of times. The spread of those values stands in for the spread across real samples.

REAL-WORLD SCENARIO

A report says a new email lifted spend by 118%. The useful question is how sure: could the true lift be 20%, or 300%? The interval answers that, but only if its recipe really covers the truth as often as it claims.

SYSTEM DIAGRAM



Top: the bootstrap from one sample. Bottom: the coverage check, where a whole group plays the population so the true value is known.

Coverage is measured, not assumed. A whole Hillstrom group (about 21,300 customers) is treated as the population, so its true rate or mean is known. Samples of n customers are drawn from it with replacement, an interval is built from each sample alone, and coverage is the share of intervals that contain the truth. Section 4 returns to Day 161's UCI Adult models; every earlier part is imported unchanged.

Today's part for the week's build

`day163_exercise.py` is today's toolkit (Wald and Wilson intervals, the bootstrap, percentile and BCa intervals, the jackknife), each checked by hand, including a bootstrap of three numbers whose 27 resamples can be listed. It is the third part of Day 167's report card.

ACTUAL OUTPUT

```
wald_ci, 5 of 100: by hand 0.05 +/- 1.96 x sqrt(0.05 x 0.95 / 100) = 0.05 +/- 0.0427 = [0.0073,
    0.0927]
  got [0.0073, 0.0927]; and for 0 of 50: [0.0, 0.0] (a zero-width interval)
wilson_ci, 0 of 50: by hand centre (0 + 3.8415/100) / (1 + 3.8415/50) = 0.0357, half-width 1.96 x
    sqrt(3.8415 / 10000) / 1.0768 = 0.0357, so [0.0000, 0.0713]
  got [0.0000, 0.0713]
bootstrap_reps of the mean of (1, 2, 3): by hand 27 equally likely resamples, mean 2, spread
    sqrt((2/3)/3) = 0.4714; a mean of 1 has chance 1/27 = 0.037, above 0.025, so the 95%
    percentile interval is [1, 3]
  got mean 1.991, spread 0.460, interval (1.0, 3.0)
jackknife of the mean of (1, 2, 3, 10): by hand 15/3, 14/3, 13/3, 6/3 = 5.0, 4.667, 4.333, 2.0;
    got [5.0, 4.667, 4.333, 2.0]
bca_ci on 1..9 (symmetric, so the acceleration is 0 and BCa should sit close to the percentile
    interval):
  percentile [3.333, 6.667], BCa [3.222, 6.556]
  BCa sits one step lower: 0.055 of resample means land exactly on 5 and only the 0.471 strictly
      below count, so its bias correction z0 is below 0
bca_ci when every value is 0: (0.0, 0.0) (a single point)
```

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. Intervals for a rate: Wald, Wilson, and the bootstrap

DEFINITION — start here, no prior knowledge assumed

The **Wald** interval is the textbook one: the observed rate plus or minus 1.96 estimated spreads. When a sample has no successes at all, the estimated spread is 0 and the interval is the single point 0.

Wilson's interval asks instead which true rates would not be rejected by a test, which pulls it towards 1/2 and keeps it wide when successes are few. The **bootstrap** of a 0/1 sample is a binomial draw at the sample's own rate, so 1,000 resamples are 1,000 binomial draws.

REAL-WORLD SCENARIO

Five people out of 100 click an ad. Is the true click rate 5% give or take 4 points, and what if nobody clicked?

MATHEMATICS

$$\text{Wald: } p \pm z\sqrt{p(1-p)/n}; \quad \text{Wilson: } [p + z^2/2n \pm z\sqrt{p(1-p)/n + z^2/4n^2}] \,/\, (1 + z^2/n); \quad z = 1.96$$

IMPLEMENTATION

```
def wilson_ci(x, n, alpha=ALPHA):
    z = NormalDist().inv_cdf(1 - alpha / 2); p = x / n
    centre = (p + z * z / (2 * n)) / (1 + z * z / n)
    half = z * np.sqrt(p * (1 - p) / n + z * z / (4 * n * n)) / (1 + z * z / n)
    return centre - half, centre + half
```

ACTUAL OUTPUT

```
population: the 21306 no-email customers; 4000 samples per size, drawn with replacement;
coverage = share of 95% intervals that contain the population's true rate (should be 0.95)
  conversion (true rate 0.0057):
    n   50: Wald 0.234, Wilson 0.975, bootstrap 0.234; samples with no successes 0.766
    n  100: Wald 0.429, Wilson 0.982, bootstrap 0.429; samples with no successes 0.570
    n  300: Wald 0.832, Wilson 0.965, bootstrap 0.825; samples with no successes 0.167
    n 1000: Wald 0.925, Wilson 0.946, bootstrap 0.918; samples with no successes 0.004
    n 3000: Wald 0.947, Wilson 0.949, bootstrap 0.939; samples with no successes 0.000
  visit (true rate 0.1062):
    n   50: Wald 0.904, Wilson 0.936, bootstrap 0.898; samples with no successes 0.003
    n  100: Wald 0.957, Wilson 0.953, bootstrap 0.950; samples with no successes 0.000
    n  300: Wald 0.951, Wilson 0.951, bootstrap 0.947; samples with no successes 0.000
    n 1000: Wald 0.945, Wilson 0.947, bootstrap 0.942; samples with no successes 0.000
    n 3000: Wald 0.953, Wilson 0.948, bootstrap 0.949; samples with no successes 0.000
Actual output from this lesson's run.
```

What it means

For purchases (true rate 0.0057) the Wald interval failed badly in small samples: coverage 0.234 at 50 customers and 0.429 at 100, because 0.570 of samples of 100 contained no buyer and gave the interval [0, 0]. The bootstrap failed the same way (0.429): resampling a sample of zeros only gives zeros. Wilson's interval stayed close to its promise at every size, between 0.936 and 0.982 (slightly too cautious at 100). From 3,000 customers all three agree (0.947, 0.949, 0.939).

For visits (true rate 0.1062) all three were close to 0.95 from 100 customers on. The textbook interval is fine for common outcomes and wrong for rare ones in small samples.

WHY CHOOSE THIS?

Wilson's interval costs nothing extra and stays honest when successes are rare or absent.

WHY NOT THIS? (TRADEOFFS)

It is slightly too wide (cautious) in small samples, and like every rate interval it assumes the rows are independent.

2. Intervals for a mean of mostly zeros

DEFINITION — start here, no prior knowledge assumed

Spend per customer is almost always zero and occasionally hundreds of dollars. Three intervals for the mean are compared: the **normal** one (mean ± 1.96 standard errors), the bootstrap **percentile** interval (the middle 95% of 1,000 bootstrap means), and **BCa** (bias-corrected and accelerated), which shifts the percentiles to allow for a bootstrap distribution that is off-centre (z_0) or whose spread changes with the value (the acceleration a, from the jackknife).

REAL-WORLD SCENARIO

Estimating average insurance claims per policy, when most policies claim nothing and a few claim a fortune.

MATHEMATICS

BCa: use the percentiles $\Phi(z_0 + (z_0 + z_q)/(1 - a(z_0 + z_q)))$ *for* $q = 0.025, 0.975$

$z_0 = \Phi^{-1}$*(share of bootstrap values below the estimate);* $a = \Sigma d^3 / (6(\Sigma d^2)^{3/2})$, *d = jackknife mean – jackknife values*

IMPLEMENTATION

```
def bca_ci(theta, reps, jack, alpha=ALPHA):
    nd = NormalDist()
    if np.all(reps == reps[0]):
        return float(theta), float(theta)
    z0 = nd.inv_cdf(min(max(np.mean(reps < theta), 1e-6), 1 - 1e-6))
    d = jack.mean() - jack; den = 6 * np.sum(d ** 2) ** 1.5
    acc = np.sum(d ** 3) / den if den > 0 else 0.0
    out = []
    for q in (alpha / 2, 1 - alpha / 2):
        zq = nd.inv_cdf(q)
        out.append(float(np.quantile(reps, nd.cdf(z0 + (z0 + zq) / (1 - acc * (z0 + zq))))))
    return tuple(out)
```

ACTUAL OUTPUT

```
population: spend of the 21307 men's-email customers: mean $1.423, 0.9875 spent nothing, largest $499
300 samples per size; each interval from 1000 bootstrap resamples
  n  200: coverage normal 0.730, percentile 0.740, BCa 0.740; mean width $4.087 / $3.764 / $4.793; samples with no buyer
      0.070 (2 s)
  n 1000: coverage normal 0.880, percentile 0.883, BCa 0.923; mean width $2.040 / $2.007 / $2.280; samples with no buyer
      0.000 (3 s)
  n 5000: coverage normal 0.933, percentile 0.940, BCa 0.953; mean width $0.970 / $0.967 / $0.997; samples with no buyer
      0.000 (21 s)
(with 300 samples, a coverage of 0.95 is measured with a spread of about 0.013)
Actual output from this lesson's run.
```

What it means

In the population 0.9875 spent nothing. With 200 customers every interval fell well short of 95% (normal 0.730, percentile 0.740, BCa 0.740): 0.070 of samples contained no buyer at all, and a sample of 200 holds 2.5 buyers on average (1.25% of 200), so it rarely sees how large spend can be. At 1,000 customers BCa was clearly best (0.923 against 0.880); at 5,000 it reached 0.953 while the other two still fell slightly short (0.933, 0.940). Each coverage is measured with a spread of about 0.013.

WHY CHOOSE THIS?

BCa corrects the skew that heavy-tailed data give a bootstrap distribution, and was the best of the three here.

WHY NOT THIS? (TRADEOFFS)

No interval can fix a sample too small to contain the rare large values; BCa also needs a jackknife (n refits).

3. The bootstrap where no simple formula exists

DEFINITION — start here, no prior knowledge assumed

A median and a ratio have no simple standard-error formula; the bootstrap does not need one. The relative lift in spend (men's email against none) is a ratio of two group means; each group is resampled on its own, because the two groups are different customers.

The bootstrap is itself random. To see how many resamples B are enough, the same interval is built 10 times with different seeds at B = 200, 1,000 and 5,000.

REAL-WORLD SCENARIO

A charity's median donation, with an interval, for its annual report; no textbook formula covers a median.

IMPLEMENTATION

```
def bootstrap_reps(stat, arrays, B=B_BOOT, seed=0):
    rng = np.random.RandomState(seed); n = len(arrays[0])
    out = []
    for _ in range(B):
        idx = rng.randint(0, n, n)
        out.append(stat(*[a[idx] for a in arrays]))
    return np.array(out)
```

ACTUAL OUTPUT

```
median spend of the 267 men's-email buyers: $66.98; 95% percentile [58.13, 83.26], BCa [57.73, 80.14]
  the bootstrap medians take only 50 different values: a median can only be one of the data's values
relative lift in spend, men's email vs none: +1.179; 95% percentile [+0.629, +1.952] (each group resampled separately, 2000
    times)
how many resamples? the median's 95% interval, 10 times with different seeds:
  B  200: lower end 57.30 to 60.62, upper end 80.15 to 83.28
  B 1000: lower end 57.73 to 58.13, upper end 80.61 to 83.32
  B 5000: lower end 57.90 to 58.13, upper end 81.64 to 83.30
Actual output from this lesson's run.
```

What it means

The men's-email buyers' median spend was $66.98, with a percentile interval [58.13, 83.26] and BCa [57.73, 80.14]. The bootstrap medians took only 50 different values: a median is always one of the data's own values, so the distribution is lumpy. A likely reason the two intervals differ, not tested here: with so many ties, BCa's bias correction (which counts only values strictly below the estimate) moves the ends; the exercise shows the same effect on a small example.

The lift in spend was +1.179 (+118%), with interval [+0.629, +1.952]: certainly positive, but anywhere from about +63% to +195%. On B: with 200 resamples the lower end wandered from 57.30 to 60.62 across seeds; with 5,000 it stayed within 57.90 to 58.13.

WHY CHOOSE THIS?

One recipe for any statistic: a median, a ratio, a model's score.

WHY NOT THIS? (TRADEOFFS)

Random itself (use thousands of resamples), lumpy for medians and quantiles, and only as good as the sample it resamples.

4. Intervals for a model's score

DEFINITION — start here, no prior knowledge assumed

Day 161's logistic regression and boosting (200 rounds) are scored on the 6,032 validation rows. Three kinds of interval: (a) **validation rows resampled**, which treats the fitted model as fixed; (b) the **gap** between the models, resampled **paired** (the same rows for both) or **unpaired** (independent rows for each); (c) **refits**: each model retrained on bootstrap resamples of the fitting rows, which measures how much the training sample alone moves the score.

REAL-WORLD SCENARIO

A team reports 86.6% accuracy for a new model against 84.1% for the old one. Is a 2.5-point gain reliable on 6,000 test rows?

ACTUAL OUTPUT

```
  logistic: accuracy 0.8408 [0.8324, 0.8498], AUC 0.8972
  boosting: accuracy 0.8662 [0.8581, 0.8748], AUC 0.9258
accuracy gap, boosting minus logistic: +0.0254; paired [+0.0192, +0.0318] (width 0.0126), unpaired [+0.0133, +0.0375] (width
    0.0242)
AUC gap, boosting minus logistic: +0.0286; paired [+0.0245, +0.0326] from 1000 resamples (63 s)
refitting: each model retrained on bootstrap resamples of the fitting rows, scored on the same validation rows
  logistic: 50 refits, validation accuracy spread 0.0010, against 0.0045 from resampling the validation rows (9 s)
  boosting: 20 refits, validation accuracy spread 0.0017, against 0.0043 from resampling the validation rows (176 s)
Actual output from this lesson's run.
```

What it means

Accuracy: logistic regression 0.8408 [0.8324, 0.8498], boosting 0.8662 [0.8581, 0.8748]. The intervals do not overlap, but the right question is the gap: +0.0254, paired interval [+0.0192, +0.0318] (width 0.0126), unpaired [+0.0133, +0.0375] (width 0.0242). Pairing halves the width, for the same reason McNemar beat the unpaired test on Day 162. The AUC gap was +0.0286 [+0.0245, +0.0326].

Refitting moved validation accuracy much less than resampling the validation rows did: a spread of 0.0010 against 0.0045 for logistic regression, 0.0017 against 0.0043 for boosting (only 20 refits, so that spread is rough). With 24,000 training rows, the test set's own randomness dominates. Boosting's training sample matters more than logistic regression's, as its higher variance on Day 161 suggested.

WHY CHOOSE THIS?

Paired resampling of a test set is cheap (no retraining) and gives the honest uncertainty of a model comparison.

WHY NOT THIS? (TRADEOFFS)

It treats the fitted models as fixed; refitting covers the training sample too, but costs one full training per resample.

5. Where the bootstrap fails: the largest value

DEFINITION — start here, no prior knowledge assumed

The bootstrap assumes the sample looks like the population in miniature. For the largest value that is never true: no resample can exceed the sample's own largest value, but the population's largest is almost always bigger. The 95th percentile, by contrast, is a stable statistic. Both are tested on last year's spend (the history column), 500 samples of 500 customers.

REAL-WORLD SCENARIO

Estimating the biggest flood a river will ever have from 50 years of records: the records cannot show a flood larger than any in them.

ACTUAL OUTPUT

```
population: last year's spend of 21307 customers; 500 samples of 500, 1000 resamples each
  largest        : true $3215.97, 95% percentile interval covered it in 0.030 of samples
  95th percentile: true $761.73, 95% percentile interval covered it in 0.940 of samples
one sample: largest $1460.57; 0.660 of its bootstrap resamples repeat that same largest value
Actual output from this lesson's run.
```

What it means

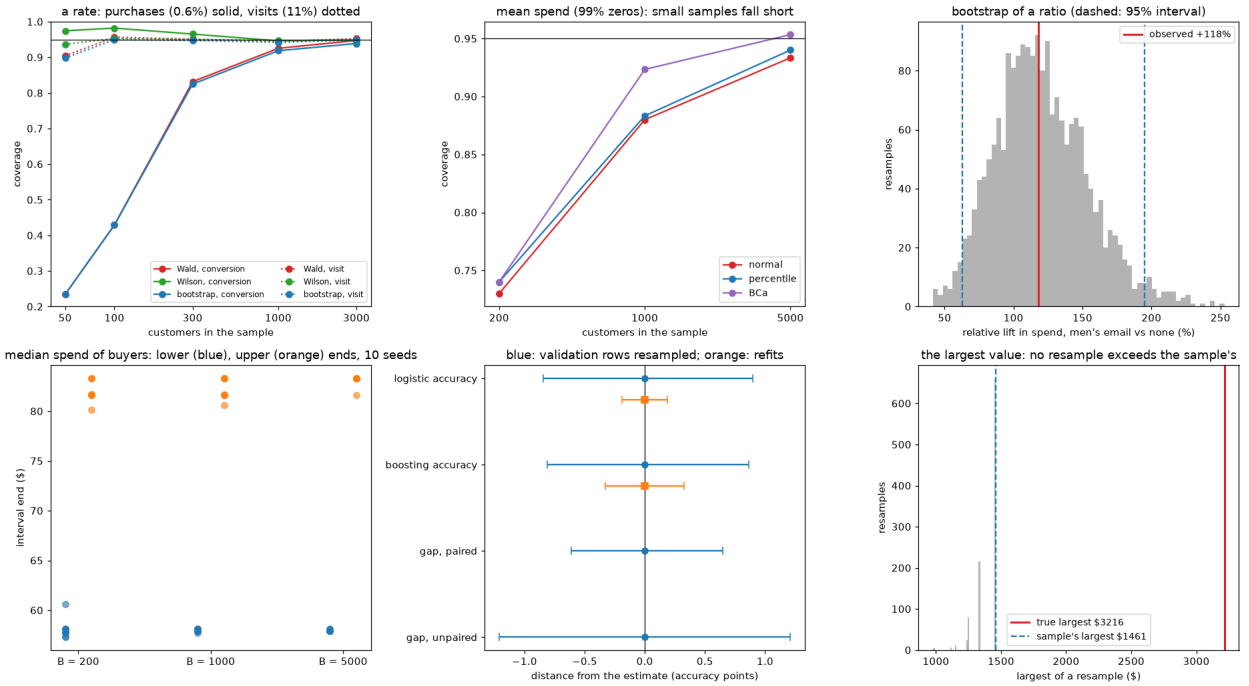
The interval for the largest value covered the true $3215.97 in only 0.030 of samples. The interval for the 95th percentile (true $761.73) covered it in 0.940. In one sample whose largest was $1460.57, 0.660 of the bootstrap resamples repeated that same value: the bootstrap distribution piles up at the sample's maximum and can never reach beyond it.

WHY CHOOSE THIS?

Knowing where the bootstrap breaks (extremes, very small samples, dependent rows) prevents confident nonsense.

WHY NOT THIS? (TRADEOFFS)

Extremes need other tools: extreme-value models, or simply reporting the sample maximum with no interval.



Top: coverage for a rare rate (solid) and a common one (dotted); coverage for mean spend; the bootstrap of the lift in spend. Bottom: interval ends from 10 seeds at three resample counts; model-score intervals drawn around their own estimates (blue: test rows resampled, orange: refits); the bootstrap of a maximum against the truth.

Interview preparation — technical questions

Q: What does 95% confidence mean?

A: The recipe contains the truth in 95% of samples. Here the Wald recipe for a rare rate managed 0.429 at 100 customers, so the label was false.

Q: Which interval would you use for a conversion rate?

A: Wilson's: it stayed between 0.936 and 0.982 at every size tested, including samples with no conversions, where Wald and the bootstrap collapse to [0, 0].

Q: How do you put an interval on a model's accuracy gap?

A: Bootstrap the test rows paired (the same rows for both models): width 0.0126 against 0.0242 unpaired.

Q: How many bootstrap resamples?

A: Thousands: with 200 the interval's lower end moved by about 3 dollars across seeds; with 5,000 by 0.23.

Q: When does the bootstrap fail?

A: For extremes: an interval for the largest value covered the truth 3% of the time. Also tiny samples and dependent rows.

Interview preparation — theoretical questions

Q: Why is a percentile interval not enough for skewed data?

A: The bootstrap distribution is itself skewed and off-centre; BCa corrects both and reached 0.953 at 5,000 customers where the percentile interval gave 0.940.

Q: What does the jackknife do in BCa?

A: Leaving out each row in turn shows how the statistic's spread changes with its value: the acceleration.

Q: Does a test set's interval cover training randomness?

A: No: it treats the model as fixed. Refitting showed training adds a spread of 0.0010 to 0.0017 here, against 0.0045 from the test rows.

Q: Why did all mean intervals fail at 200 customers?

A: 7% of samples had no buyer and most had only a few: the sample had not seen the rare large purchases that drive the mean.

Key takeaway and what's next

> **KEY TAKEAWAY**
>
> An interval's 95% is a property of its recipe, and recipes can be checked. The textbook interval failed for rare events in small samples, every interval failed for heavy-tailed means in small samples, and the bootstrap failed for extremes; Wilson's interval and BCa held up best. For models, paired resampling gives the honest width of a comparison.

- Rare rate at 100 customers: Wald 0.429, Wilson 0.982.
- Mean spend: coverage 0.730 at 200 customers; BCa 0.953 at 5,000.
- Accuracy gap +0.0254: paired width 0.0126, unpaired 0.0242.
- Largest value: covered in 0.030; 95th percentile 0.940.

Suggested watch order, if starting from zero

- 1. Confidence Intervals, Clearly Explained!!! (StatQuest with Josh Starmer) — confidence intervals built by bootstrapping, with pictures.
- 2. The standard error, Clearly Explained!!! (StatQuest with Josh Starmer) — the standard error, and how bootstrapping estimates it.

Day 164 preview

- **The multiple-comparisons problem.** Hillstrom's customers are cut into many segments and each is tested; some come out 'significant' by chance alone. Bonferroni, Holm and Benjamini-Hochberg are built and checked on data where the truth is known.